

Exercise No 12

Q1.Develop a geometric dimensions tool that stores the fixed height, width, and depth parameters of a shipping container inside a tuple, unpacks the measurements, and computes the total volume.

Code:

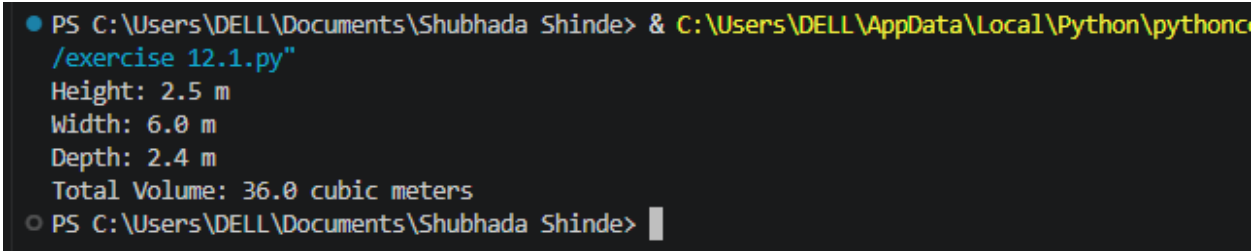
```
dimensions = (2.5, 6.0, 2.4)

height, width, depth = dimensions

volume = height * width * depth

print("Height:", height, "m")
print("Width:", width, "m")
print("Depth:", depth, "m")
print("Total Volume:", volume, "cubic meters")
```

Output:



```
● PS C:\Users\DELL\Documents\Shubhada Shinde> & C:\Users\DELL\AppData\Local\Python\python.exe /exercise 12.1.py
Height: 2.5 m
Width: 6.0 m
Depth: 2.4 m
Total Volume: 36.0 cubic meters
○ PS C:\Users\DELL\Documents\Shubhada Shinde>
```

Q2. Create a constant system logging profile that maps an IP address alongside a server port number inside a tuple pair, demonstrating that these settings remain secure from runtime modification attempts.

Code:

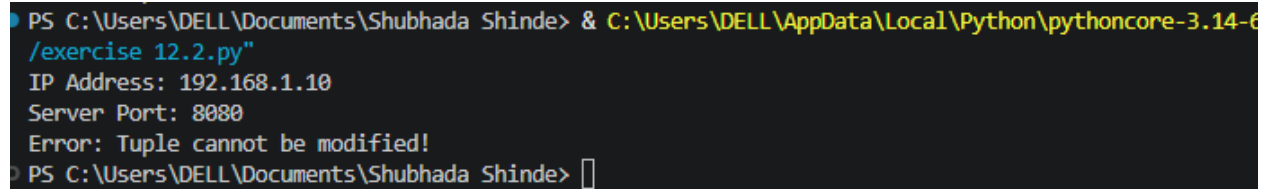
```
logging_profile = ("192.168.1.10", 8080)

ip_address, port = logging_profile

print("IP Address:", ip_address)
print("Server Port:", port)

try:
    logging_profile[0] = "192.168.1.20"
except TypeError:
    print("Error: Tuple cannot be modified!")
```

Output:



```
PS C:\Users\DELL\Documents\Shubhada Shinde> & C:\Users\DELL\AppData\Local\Python\pythoncore-3.14-64\python.exe exercise_12.2.py
IP Address: 192.168.1.10
Server Port: 8080
Error: Tuple cannot be modified!
PS C:\Users\DELL\Documents\Shubhada Shinde>
```